Ans to the Q. no-1

$X = (9.321)_{10}$

$= (1001.0101)_2$

$Y = (3.325)_{10}$

$= (11.0101)_2$

Normalizing both X and Y we get,

$X = 1.0010101 \times 2^3$

$Y = 1.10101 \times 2^1$

Now, $X * Y$

$= (1.0010101 \times 2^3) \times (1.10101 \times 2^1)$

$= (1.0010101 \times 1.10101) \times 2^4$

$= 1.111011011001 \times 2^4$

(Ans)

$0.321 \times 2 = 0.642$

$0.642 \times 2 = 1.284$

$0.284 \times 2 = 0.568$

$0.568 \times 2 = 1.136$

$0.325 \times 2 = 0.65$

$0.65 \times 2 = 1.3$

$0.3 \times 2 = 0.6$

$0.6 \times 2 = 1.2$

Ans to the Q: no-2

$18.7110 - (-7.0110) = 18.7110 + 7.0110$

$(18.7110)_{10} = (10010.1011)_2$

$(7.0110)_{10} = (111.0000)_2$

Normalizing both numbers we get,

$(10010.1011)_2 = 1.00101011 \times 2^4$

$(111.0000)_2 = 1.11 \times 2^2$

Now, $(1.00101011 \times 2^4) + (1.11 \times 2^2)$

$= (1.00101011 \times 2^4) + (0.0111 \times 2^4)$

$= (1.00101011 + 0.0111) \times 2^4$

$= 1.10011011 \times 2^4$

(Ans)

$0.7110 \times 2 = 1.422$

$0.422 \times 2 = 0.844$

$0.844 \times 2 = 1.688$

$0.688 \times 2 = 1.376$

$0.0110 \times 2 = 0.022$

$0.022 \times 2 = 0.044$

$0.044 \times 2 = 0.088$

$0.088 \times 2 = 0.176$

$\text{Bias} = 2^{8-1} - 1 = 127$

$\text{Biased Exponent} = 127 + 4 = 131$

$\text{Range of Bias exponent} = 0 \text{ to } 2^8 - 1$

$= 0 \text{ to } 255$

$= 1 \text{ to } 254$

$1 < 131 < 254$

No overflow or underflow